

Coding & Solution

Date	26 June 2026
Team ID	N/A
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/danduthanishkareddy/Voice-Based-Concept-Understanding-Analyser
Programming Language(s)	Python
Framework(s) Used	Streamlit

Key Features Implemented	Audio upload, audio playback, waveform visualization, speech-to-text transcription, semantic similarity calculation, filler word ratio, audio feature extraction, understanding level prediction, and downloadable PDF report generation.
Pending / Incomplete Features	User authentication, database storage, live microphone recording, and cloud deployment can be added in future.
Setup / Run Instructions	Install required libraries, open the project folder in VS Code, run <code>python -m streamlit run app.py</code> , upload an audio file, click Analyze Concept Understanding, and download the PDF report.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Partial
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Voice Based Concept Understanding Analyser is implemented using a modular Python and Streamlit-based structure. The application successfully accepts audio input, processes speech, evaluates conceptual understanding using semantic similarity, analyzes basic audio features, and generates a PDF report. The code is organized into separate modules for transcription, semantic evaluation, audio utilities, scoring, and report generation, which improves readability and reusability. Future enhancements can include login functionality, database storage, multilingual support, and deployment on Streamlit Cloud.
--